SMART STUDY PLANNER

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INTRODUCTION:

In today's academic environment, students often struggle with time management and performance tracking. Many students prepare for examinations without systematically tracking their study hours or analyzing their subject-wise performance. As a result, they are unable to identify weak subjects or measure improvement over time.

This project presents a Smart Study Planner & Performance Analytics System developed using Python. The system allows students to log study hours, record test scores, analyze performance trends, and predict future results using basic data analysis techniques.

PROBLEM STATEMENT:

Students lack a simple and effective tool to:

- Track daily study hours
- Monitor subject-wise performance
- Identify weak subjects
- Analyze improvement trends

Manual tracking is inconsistent and does not provide data-driven insights.

OBJECTIVES:

The main objectives of this project are:

- To create a system that logs study sessions.
- To record subject-wise test scores.
- To analyze total study hours per subject.
- To calculate average performance percentage
- To identify the weakest subject.
- To predict future performance using regression.

SCOPE OF THE PROJECT:

This project is designed for:

- School and college students.
- Self-learners preparing for competitive exams.
- Students who want data-driven academic improvement.

This system is a console-based Python application and can be extended into a web or mobile application in the future.

TOOLS AND TECHNOLOGIES USED:

- Python 3
- Pandas (Data Handling)
- NumPy (Mathematical Computation)
- Matplotlib (Data Visualization)
- CSV Files (Data Storage)
- Git & GitHub (Version Control)

SYSTEM ARCHITECTURE:

The system consists of four main modules:

- Data Handler Module
  - Adds study sessions
  - Adds test score
- Analytics Module
  - Calculate total study hours
  - Computes average percentages
  - Identifies weak subjects
  - Generates graphs
- Predictor Module
  - Uses linear regression
  - Predicts next test score
- Main Module
  - Provides user interface
  - Controls program flow

METHODOLOGY:

- STEP 1: Data Collection

Users enter study hours and test scores.

- STEP 2: Data Storage

Data is stored in CSV files inside a structured project directory.

- STEP 3: Data Processing

Pandas is used to group, calculate averages, and compute percentages.

- STEP 4: Visualization

Matplotlib generates bar graphs to display study and performance trends.

- STEP 5: Prediction

NumPy's function performs simple linear regression to predict future scores.

IMPLEMENTATION DETAILS:

The project follows modular programming principles:

- data_handler.py manages input and CSV storage.
- analytics.py performs calculations and visualizations.
- predictor.py handles regression-based prediction.
- main.py provides menu-driven user interaction.

The system uses dynamic path handling to avoid file errors across systems.

SAMPLE OUTPUT:

Example Menu:

Smart Study Planner

- Add Study Session
- Add Test Score
- Study Analysis
- Score Analysis
- Weak Subject
- Predict Next Score
- Exit

Example Output:

Weakest Subjects: Physics

Predict next score for Mathematics: 85.2

Graphs display subject-wise study hours and average percentages.

CHALLENGES FACED:

- File path errors while reading CSV files.
- Handling missing or insufficient data for prediction.
- Managing Git installation and version control.
- Ensuring modular structure and clean code.

All issues were resolved through debugging and improved file path management using os. path.

LEARNING OUTCOMES:

Through this project, I learned:

- Practical application of python in real-world problem solving.
- Data manipulation using Pandas.
- Data visualization using Matplotlib.
- Basic machine learning concept (Linear Regression).
- Drbugging and file path handling.
- Git and GitHub version control.
- Writing structured documentation.

EVALUATION CRITERIA ALIGNMENT:

- Relevance to Course Concepts

The project applies programming, data analysis, and visualization concepts learned in the course.

- Code Quality

The project uses modular structure and clear documentation.

- Reflection Depth

Challenges and learning experiences are clearly discussed.

- Usability

The README file explains installation and usage clearly.

FUTURE SCOPES:

The project can be enhanced by:

- Converting into a Web Application using Flask.
- Adding a graphical user interface (GUI).
- Integrating AI-based study recommendations.
- Adding authentication and multi-user support.
- Deploying a cloud platform.

CONCLUSION:

The Smart Study Planner & Performance Analysis System successfully addresses a real-world student problem. The system enables structured study tracking, data-driven performance analysis, and predictive insights. The project demonstrates the practical implementation of programming and analytical concepts learned during the course.

It highlights how simple technological solutions can significantly improve academic planning and performance monitoring.

RESOURCES/REFRENCES:

- Official Python Documentation – https://docs.python.org/3/
- Pandas Documentation – https://pandas.pydata.org/docs/
- Matplotlib Documentation – https://matplotlib.org/stable/index.html
- Git Documentation – https://git-scm.com/docs
- GitHub Guides – https://guides.github.com/
- W3Schools Python Tutorial – https://www.w3schools.com/python/
- GeeksforGeeks Python Tutorials – https://www.geeksforgeeks.org/python-programming-language/